23/7/26

EXPERIMENT-3

TO STUDY THE PACKET TRACER TOOL INSTALLATION AND USER INTERFACE OVERVIEW

AIM:

To understand environment of CISCO PACKET TRACER to design simple network.

INTRODUCTION:

A simulator, simulates network devices and its environment. Packet Tracer is an exciting network design, simulation & modelling tool.

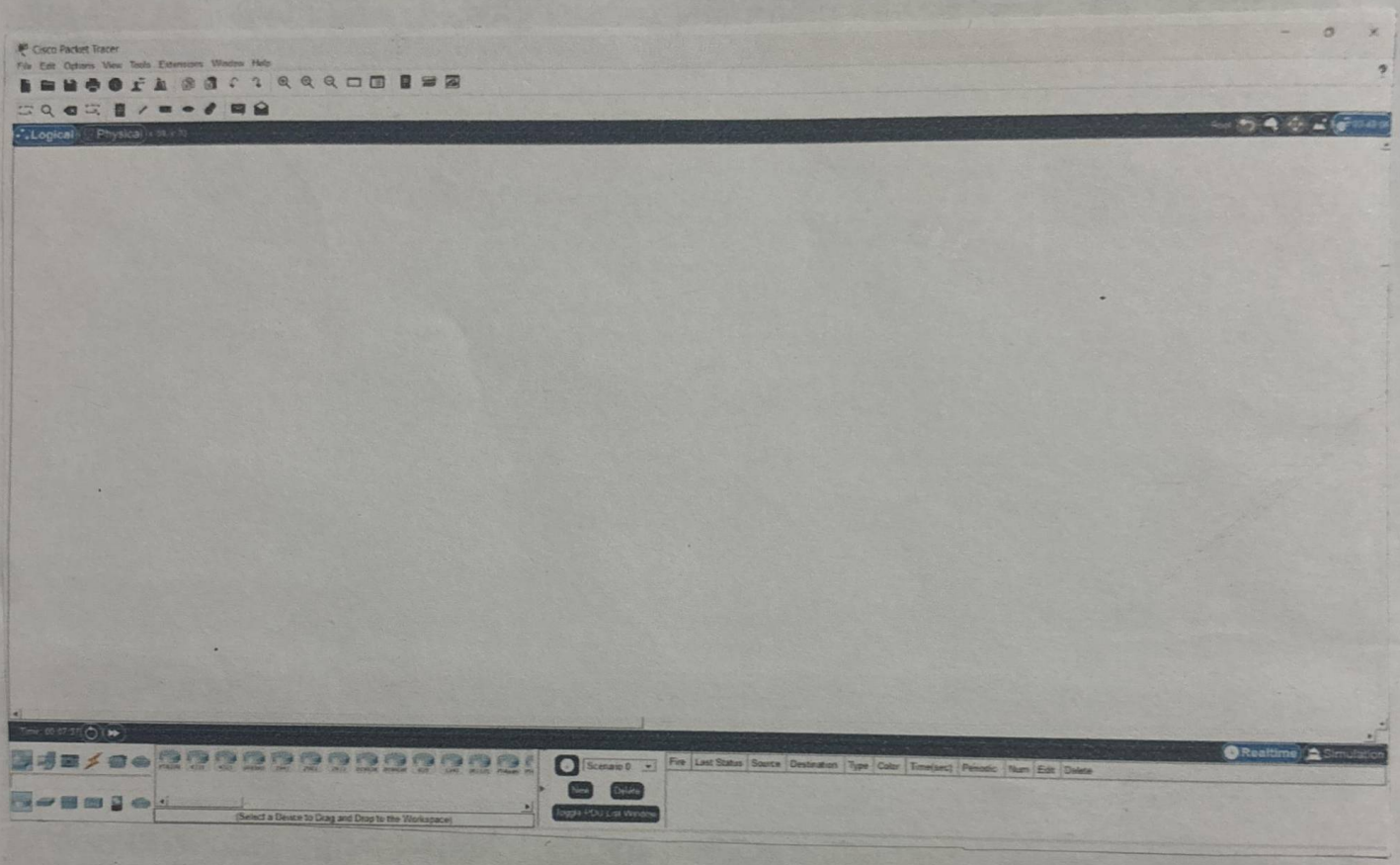
1. It allows you to model complex systems without the need for dedicated equipment.
2. It helps you to practice your network configuration and troubleshooting skills via computer or an Android or iOS based mobile device.
3. It is available for both the Linux & Windows desktop environments.

INSTALLING PACKET TRACER:

To download Packet Tracer, go to https://www.netcad.com & log in with your Cisco Networking Academy credentials, then click on the Packet Tracer graphic & download the package appropriate for your OS.

WINDOWS:

Installation in Windows is pretty simple & straightforward; the setup comes in a single file named Packettracker-setup6.0.1.exe.
Open the file to begin the setup, accept the license agreement, choose a location & start the installation.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Realtime
Simulation
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)

LINUX:

Linux users with an Ubuntu/Debian distribution should download the file for Ubuntu, & those using Fedora/Redhat/CentOS must download the file for Fedora.

chmod +x PacketTracer601-i386-installer-rpm.bin
./PacketTracer601-i386-installer-rpm.bin

USER INTERFACE OVERVIEW:

The layout of PacketTracer is divided into several components. The components of the Packet Tracer interface as follows: match the numbering with explanation.

1. Menu Bar - This is common menu found in all software applications; it is used to open, save, print, change preferences & so on.

2. Main Toolbar - This bar provides shortcuts icons to menu options that are commonly accessed such as open, save, zoom, undo & redo and no the right-hand side is an icon for entering network information for the current network.

3. Logical/Physical workspace tabs - These tabs allow you to toggle b/w the logical & physical work areas.

4. Workspace - This is the area where topologies are created & simulations are displayed.

5. Real-time / Simulation Tabs - These tabs are used to toggle b/w the real & simulation modes. Buttons are also provided to control the time, to capture the packets.

6. Network component box - This component contains all of the network end devices & is further divided into 2 areas. 7a) Device-type selection box. 7b) Device-specific selection box.

ANALYSE THE BEHAVIOUR OF NETWORK DEVICES USING CISCO PACKETS TRACER SIMULATOR:

1) From the network component box, click and drag the below components:

a) 4 Generic PC's & one HUB

b) 4 Generic PCs & one SWITCH

2) Click on connections:

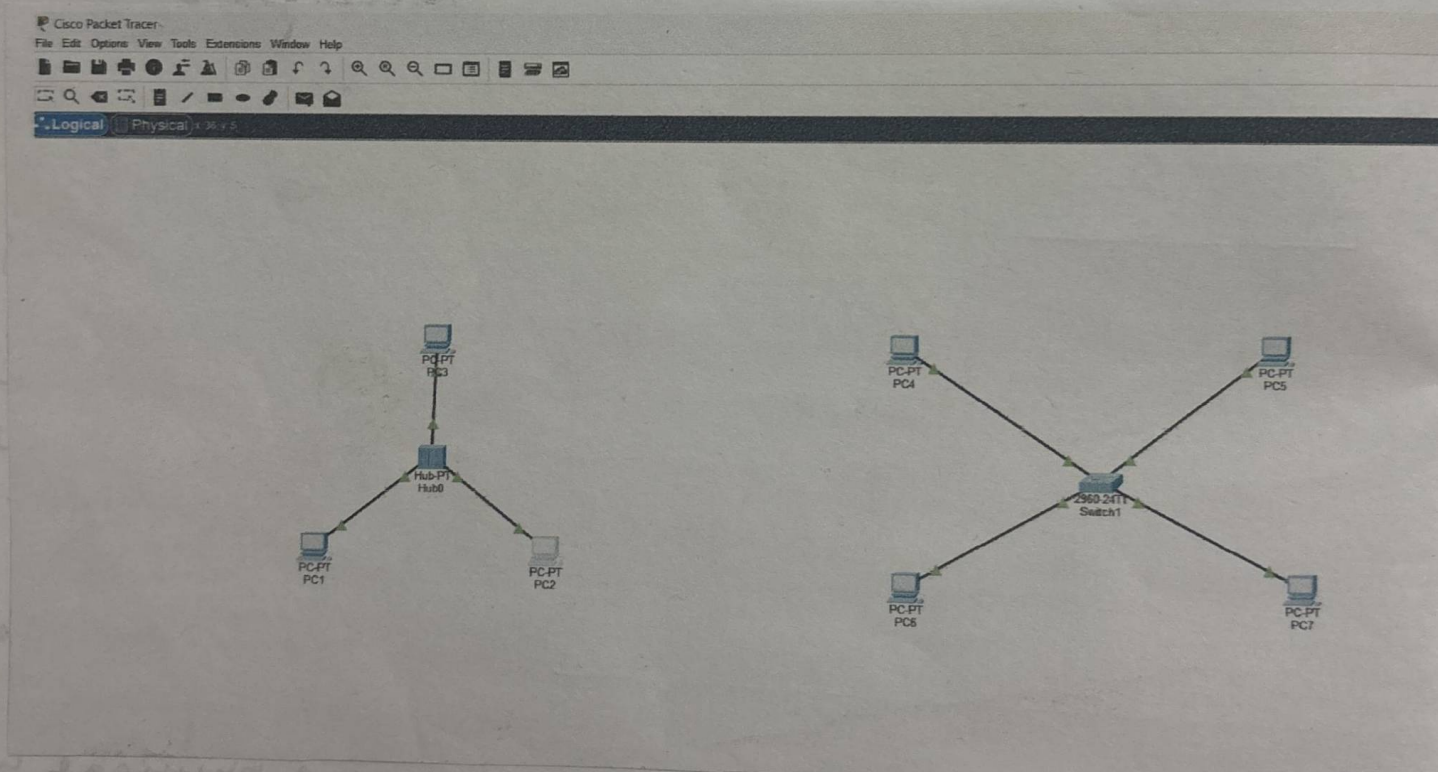
a) Click on Copper Straight-Through Cable.

b) Select one of the PC & connect it to HUB using the cable.

c) Similiarly connect 4 PC's to the switch using copper straight-through cable.

3) Click on the PCs connected to hub, go to the Desktop tab, click on IP configuration & enter IP address.

Click on PDU from the common bar.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
PC-PT PC3
Hub-PT Hub0
PC-PT PC1
PC-PT PC2
PC-PT PC4
PC-PT PC5
2960-24TT Switch1
PC-PT PC6
PC-PT PC7

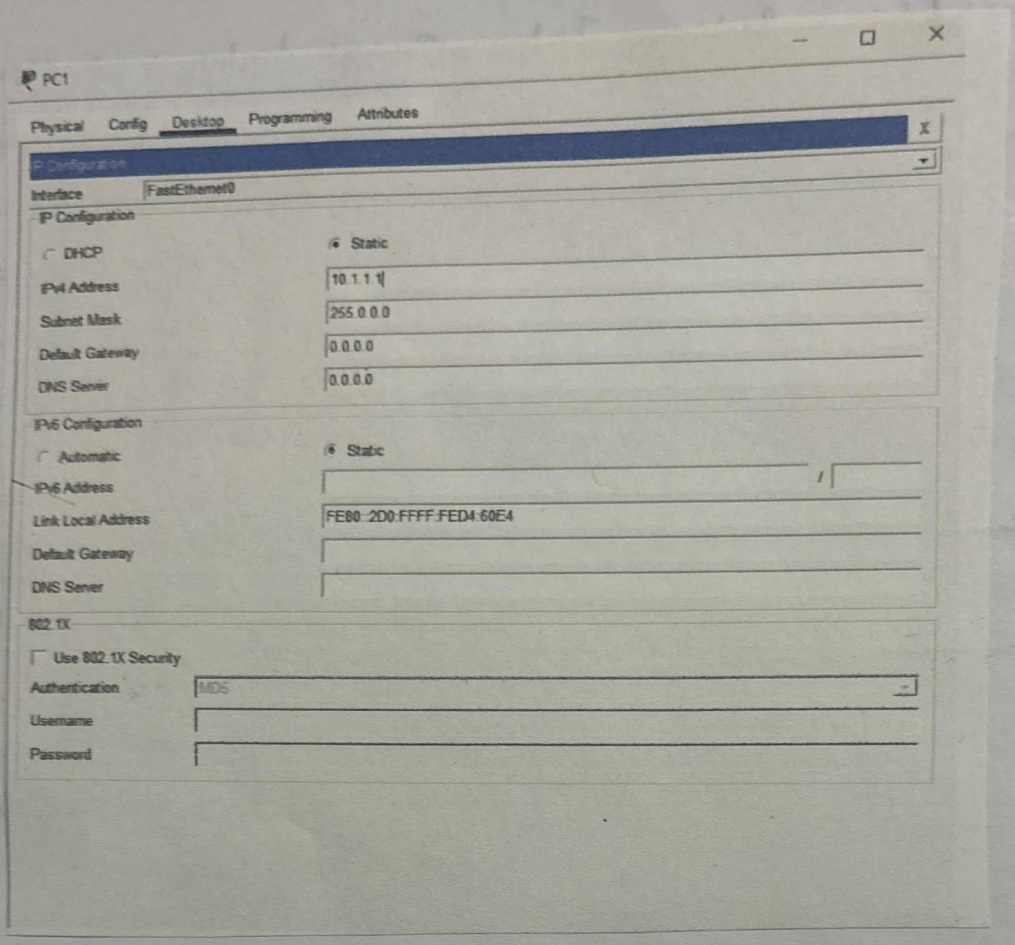
4) Observe the flow of PDU from source PC to destination PC by selecting realtime mode of simulation.

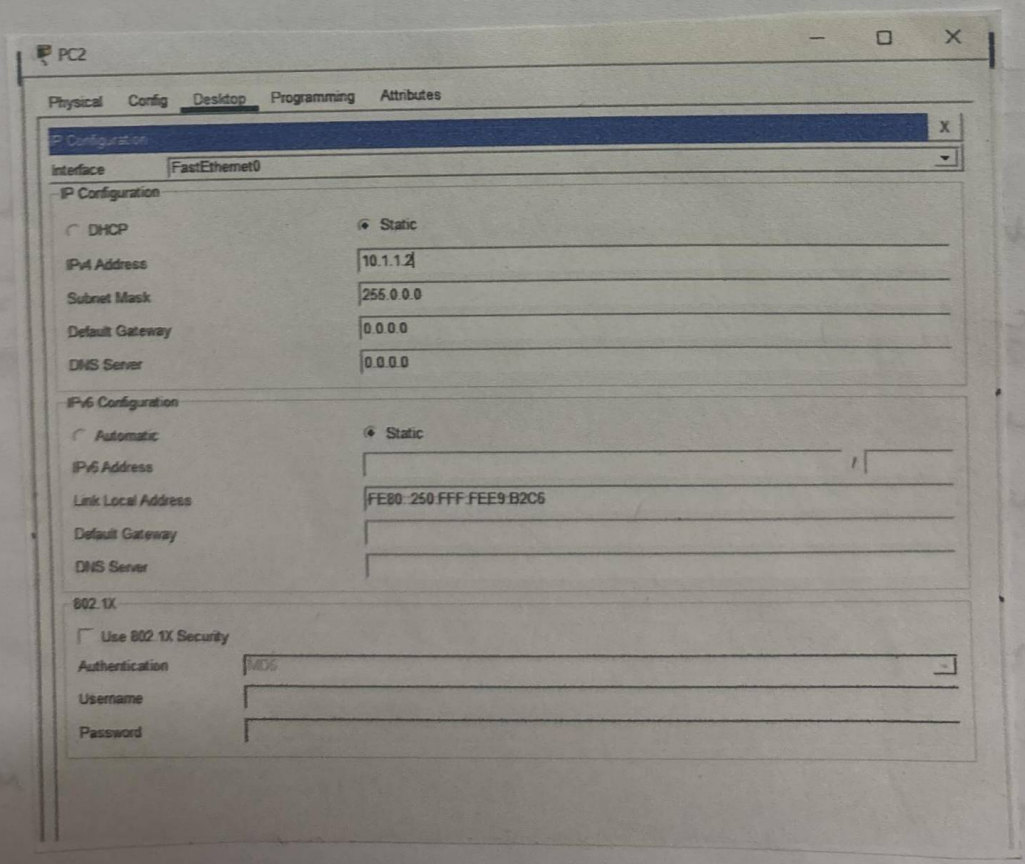
5) Repeat step #3 to step #5 for the PC's connected to the switch.

6) Observe how the HUB & switch are forwarding the PDU & write your observation & conclusion about the behaviour of switch & HUB.

RESULT:

Thus, the CISCO Packet Tracer Installation and design of simple network using hub & switch has been studied and executed successfully.


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.1.1.1
Subnet Mask
255.0.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:FFFF:FED4:60E4
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.1.1.2
Subnet Mask
255.0.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::250:FFF:FEE9:B2C6
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password